Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
void reverse(int n)
{
    int rev = 0, remainder;
    int count=0;
    count++;
    count++;
    while (n != 0)
     {
         count++;
         remainder = n % 10;
         count++;
         rev = rev * 10 + remainder;
         count++;
         n/= 10;
         count++;
     }
     count++;
//print(rev);
printf("%d",count);
}
int main()
{
    int n;
    scanf("%d",&n);
    reverse(n);
}
```

	Input	Expected	Got	
✔	13	11	11	✔
✔	1234	18	18	✔

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
void function(int n)
{
    int c= 0;
    int count=0;
    count++;
    for(int i=n/2; i<n; i++){
        count++;
        for(int j=1; j<n; j = 2 * j){
            count++;
            for(int k=1; k<n; k = k * 2){
                count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
}
int main()
{
    int n;
    scanf("%d",&n);
    function(n);
}
```

	Input	Expected	Got	
✔	4	38	38	✔
✔	10	212	212	✔

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
        printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int count=0;
void Factor(int num)
 {
    for (int i = 1; i <= num;++i)
    {
        count++;
        count++;
     if (num % i== 0)
        {
            count++;
         //printf("%d ", i);
        }

     }
     count++;
     printf("%d",count);
  }

int main()
{
    int n;
    scanf("%d",&n);
    Factor(n);
}
```

	Input	Expected	Got	
✔	13	31	31	✔
✔	25	54	54	✔
✔	6	12	12	✔

Passed all tests! ✔

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
   if(n==1)
   {
    printf("*");
   }
   else
   {
    for(int i=1; i<=n; i++)
    {
      for(int j=1; j<=n; j++)
      {
         printf("*");
         printf("*");
         break;
      }
    }
  }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
void func(int n)
{
    int count=0;
    if(n==1)
    {
      //printf("*");
      count++;
    }
    else
    {
       count++;
     for(int i=1; i<=n; i++)
     {
         count++;
         count++;
       for(int j=1; j<=n; j++)
       {
          count++;
          count++;
          //printf("*");
          //printf("*");
          break;
       }
       count++;
     }
     count++;
   }
   printf("%d",count);
}
int main()
{
    int n;
    scanf("%d",&n);
    func(n);
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
    int i=1;
    count++;
    int s=1;
    count++;
    while(s<=n){
        count++;
        i++;
        count++;
        s+=i;
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	6	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.